

ControlPlane

The real-time oversight gateway for enterprise AI

Every AI response, inspected before delivery — quality, cost, and responsibility converged into one policy-governed verdict per business pipeline. Working prototype, measured evidence, auditable claims.

Business Proposal — Accenture Innovation Challenge 2026, Round 2 (Prototype Development)

Team b24bb1029 — Indian Institute of Technology Jodhpur		
Priyanshu Pandey (Lead)	B24BB1029	Bioengineering, 3rd year
Harshit Saini	B24CS1031	Computer Science, 3rd year
Jayant Soni	B24CM1033	Artificial Intelligence & Data Science, 3rd year

Prepared 30 August 2026. Every product figure in this document is measured, reproduces from a single command in the public repository, and is presented with its limitations. Market figures are bottom-up models with stated assumptions, marked [model].

1 · Executive summary

Enterprises are deploying generative AI faster than they can supervise it. Quality is evaluated in offline reports weeks after the fact; cost appears on a monthly invoice; privacy and safety failures surface in audits — all **after the damage**. The tooling market sells point solutions for each symptom; nothing sits in the delivery path making a single, accountable decision per response, per business context.

ControlPlane is that missing layer: a drop-in gateway (one base-URL change) through which every AI request and response passes. Deterministic scanners, small ML classifiers, and grounding checks emit multi-label risk signals; a single policy engine converges them into one of four verdicts — **Pass, Edit, Block, Escalate** — under a per-pipeline policy file. Three business pipelines can receive three different verdicts on the identical content with zero code difference: governance is configuration, versioned and audited.

This is not a concept deck. The prototype is a working system with an unusual property for this stage: **its claims are auditable**. A frozen, independently reviewed 280-case evaluation corpus; blind first-contact measurements published before any tuning; misses reported next to passes; 1,200+ tests green on two machines; and an engineering ledger of 32 formally adjudicated deviations — produced by AI coding agents operating under the same class of oversight the product implements.

840/840 policy-engine verdicts correct, frozen corpus × 3 policies	1.000 PII detector precision — zero false alarms across corpus	<1 ms tier-1 hold P99 (0.77 ms); full-pipeline holds within targets	39/39 fault-injection assertions; fail-open/fail-closed per policy	1,200+ tests, two-machine certification, CI on 3.12/3.14
--	--	--	--	--

Why now: the EU AI Act's obligations are in force and phasing in; India's DPDP Act exposes enterprises to penalties up to INR 250 crore per category of security-safeguard failure; sector regulators (RBI, SEBI, HIPAA regimes) increasingly expect demonstrable AI oversight. Compliance evidence — the audit trail ControlPlane produces natively — is becoming a purchasing requirement, not a feature.

The opportunity: occupy the converged pre-delivery decision layer before point tools grow into it. Open-core distribution (Apache-licensed gateway) drives adoption; a paid Governance Console, compliance policy packs, and audit retention monetize the enterprises that need evidence. We seek hackathon backing now and an indicative pre-seed of INR 1.6 crore (~US\$190k) to convert the prototype into three design-partner pilots within two quarters.

2 · Problem framing

2.1 · Three failures, one root cause

A bank runs three AI assistants. The customer chatbot confidently invents a loan approval (**wrong**). The internal assistant repeats an employee's phone number into a log (**leaky**). The decision-support tool quietly burns tens of thousands in inference (**expensive**). Three different teams discover three different incidents — weeks late. The root cause is common: **no supervision exists at the moment the AI speaks**.

2.2 · Why current tooling does not solve it

Tool category	What it does	What it cannot do
Observability / evals (Arize, Langfuse, LangSmith, Datadog LLM)	Watches, scores, and reports after delivery	Cannot stop, edit, or escalate a response before a user sees it
Guardrails (NVIDIA NeMo, AWS Bedrock Guardrails, Azure Content Safety, Lakera)	Filters specific harms with stacked allow/deny checks	One global configuration; no per-pipeline governance, no converged verdict, no fail-posture policy
AI gateways (Portkey, Kong AI Gateway, LiteLLM)	Routing, caching, cost tracking, basic guardrail hooks	Decision logic is thin; risk categories siloed; no audit-grade evidence trail
DLP (Microsoft Presidio, Google DLP)	Broad PII category coverage	Not response-path decision-making; no quality/cost planes; no HITL loop

Each category is real and funded — and each optimizes one plane. The enterprise problem is the **convergence**: one accountable verdict per response, tuned per business context, with evidence a regulator accepts.

2.3 · The regulatory forcing function

Regime	Exposure	Operational implication
EU AI Act (in force, obligations phasing through 2026-27)	Fines up to €35M / 7% of global turnover for the most serious violations	High-risk deployments must demonstrate oversight, logging, and human control — an audit-trail requirement
India DPDP Act 2023 (rules operationalizing)	Up to INR 250 crore per category of failure of security safeguards	AI channels that touch personal data need demonstrable interception and minimization
Sectoral (RBI/SEBI expectations, HIPAA, internal model-risk regimes)	License, audit, and reputational exposure	Per-jurisdiction, per-use-case policy variance — hard-coded rules age immediately

ControlPlane's design premise follows directly: **policies are data, not code** — scoped by pipeline, geography and risk appetite, versioned, hot-reloadable, and stamped into every audit record.

3 · The solution — and what the prototype already proves

3.1 · Architecture in one paragraph

Applications change one base URL; every request then flows: **input lane** (PII scan with pre-dispatch prompt redaction, injection screen, budget gate) -> provider dispatch (two-tier model routing; provider accounting sanity-checked at boot) -> **sentence-buffered output lane** (a flagged sentence never partially reaches the user) -> detectors emit **multi-label signals** (a fabricated fact about a person is one signal carrying both a hallucination and a privacy label) -> the **policy engine** converges everything into one verdict — Pass / Edit / Block / Escalate — under the pipeline's YAML policy -> append-only **audit record** (categories always, raw values never) -> escalations enter a human review queue whose decisions feed proposed threshold updates. Uncertainty handling and even **failure posture** (fail-open vs fail-closed when a detector itself breaks) are per-pipeline policy fields.

3.2 · Application cases implemented in the prototype

Pipeline (shipped policy)	Enterprise application case	Demonstrated behaviors (all live)
support_bot — customer-facing, streaming, moderate strictness	Customer-service deflection for consumer businesses	PII redacted from prompts pre-dispatch (provider never sees it) and from streamed answers; typed credentials blocked outright (an incident, not a field); injection blocked at zero token cost; borderline claims softened
hr_copilot — internal, strict on personal data	Internal knowledge / HR assistant	Identical PII content that support_bot edits is blocked here — policy-only divergence; logged-pass posture for low-confidence claims
finance_advisor — regulated, non-streaming, fail-closed	Decision support in a regulated workflow	Full-response buffering (nothing precedes the verdict); escalation-heavy posture; quarantine + human review with released/rejected lineage; detector failure escalates rather than silently blocking

The signature demonstration — the same request producing **Edit / Block / Escalate** across the three pipelines with zero code difference — runs live and is enforced by a structural test: use-case names may not appear in executable code.

3.3 · How AI enables the solution

AI is on both sides of the gateway. **Watched:** any OpenAI-compatible provider (cloud or local — the demo runs on Groq-hosted open models and a fully local Ollama fallback). **Watching:** a deliberately tiered stack — microsecond deterministic scanners derived from published format specifications; quantized transformer classifiers (injection, toxicity) on CPU with full-coverage windowing that cannot be bypassed by padding; embedding-based grounding checks; NER-driven privacy enrichment. No LLM sits in the decision path: the verdict layer is deterministic, explainable, and testable — the correct property for a governance product.

4 · Novel systems built in this prototype

Six mechanisms we did not find combined — in several cases not present at all — in the surveyed landscape:

Mechanism	What is new	Why it matters commercially
Converged tri-plane verdict engine	Quality, cost, and responsibility signals merge into one accountable verdict; risk labels are multi-label because real risks overlap	One throat to choke for compliance; stacked-filter competitors cannot express overlapping risk
Uncertainty and failure as configuration	The borderline band's action and each detector-class's fail-open/fail-closed posture are per-pipeline policy fields	Answers the two questions regulators actually ask — what do you do when unsure, and when your safety net breaks
Attribution-correct enforcement	Budgets clock detector-attributable compute, not event-loop wall-clock; scheduling noise cannot fabricate faults; hang backstop kept separately	Verdicts in regulated pipelines cannot be flipped by load artifacts — discovered and fixed by measurement, on the record
Provenance-classed providers + accounting canary	Upstreams are dev- or measured-class; a boot canary checks provider token accounting against a local estimate	Caught a real intermediary silently injecting ~5,000 hidden prompt tokens per request — cost-integrity as a shipped feature
Evidence pipeline as product	Frozen reviewed corpus; blind-then-disclosed measurement; every README claim maps to a reproducing command; misses published beside passes	Competitors ask to be trusted; ControlPlane ships the apparatus for auditing itself — the core enterprise objection, answered structurally
Built under the oversight it sells	AI agents developed the system under a binding contract: 32 adjudicated deviations, independent AI review, human rulings — the full ledger is public	A defensible story no incumbent can copy quickly, and living proof the governance model works on real engineering

5 · Evidence — measured, reproducible, and honest about misses

Quantity	Measured	Reading
Policy-engine conformance	840/840 verdicts (280 frozen cases × 3 policies); falsified via 7/7 injected defects caught	The decision layer is provably correct against its specification
PII detection	Precision 1.000; recall 0.836 blind -> 0.885 after a disclosed, spec-derived revision (target 0.95 — missed)	Zero false alarms; every residual miss is one documented scope trade-off; both numbers ship permanently
Unsourced-numeric claims	Precision 0.267 blind -> 0.857 after correcting a flaw in our own spec	First-contact honesty: the bad number is preserved beside the fix
Injection / toxicity classifiers (blind)	Injection P 1.000, R 0.150; toxicity high P 0.400, R 0.250 — no targets were set, nothing was tuned	Layered defense: patterns catch overt attacks in microseconds; the model layer's thresholds tune via the feedback loop on real traffic — a roadmap item, stated as such
End-to-end verdict accuracy	0.825–0.851 over 194 covered cases (up from 159)	The score fell as coverage widened into the hardest classes — reported exactly that way
Latency	Input-hold P99 26.7 ms / 50; sentence-hold P99 31.4 ms / 100; tier-1 era hold P99 0.77 ms; one detector tail breach (25.35 vs 25 ms) published as a standing limitation	Budgets are enforced on attributable time; the one miss is published, target unmoved
Reliability	39/39 fault-injection assertions on a quiet host; fail-closed never silently blocks	Failure posture is policy, and it is tested

Quantity	Measured	Reading
Engineering base	1,200+ tests; CI (Python 3.12/3.14); two-machine certification; 32 adjudicated deviations; 8 standing limitations, each explained	The limitation register is the roadmap — measured, not imagined

Sources: reports/ directory of the public repository; every row reproduces from one command (README claims table).

6 · Target users and buyers

Persona	Pain today	What ControlPlane gives them
Head of Platform / AI Engineering	N applications × M models × ad-hoc guardrails; every new use case re-solves oversight	One gateway, one policy schema; onboarding a pipeline = one YAML file + one header
CISO / DPO / Compliance officer	Cannot evidence AI oversight to auditors; DPDP/AI-Act exposure	Per-jurisdiction policy packs; append-only audit trail with categories-not-values; HITL lineage
FinOps / Engineering finance	Inference spend unattributed and unmanaged	Per-pipeline budgets enforced pre-dispatch; tiered routing; cost attribution per verdict
Risk / model-governance teams (BFSI)	Regulator expects human-in-the-loop and model supervision	Escalation queue, fail-closed postures, versioned policy diffs as change evidence

Initial ICP: mid-to-large enterprises running 3+ generative-AI use cases in regulated or brand-sensitive contexts — BFSI, healthcare, telco, and India's Global Capability Centers, where DPDP exposure meets aggressive AI adoption. The design-partner motion starts there (§8).

7 · Market analysis

7.1 · Category and timing

ControlPlane sits at the intersection of AI gateways, guardrails/AI-security, and AI governance/GRC — the space industry analysts group under AI trust, risk and security management. Every serious survey of enterprise generative-AI adoption reports the same pattern: adoption is broad, production incidents are rising, and governance maturity lags years behind deployment. Precise third-party market-size figures vary by an order of magnitude across reports; rather than cite one, we model the opportunity bottom-up with stated assumptions.

7.2 · Bottom-up sizing [model — all assumptions stated]

Layer	Assumption	Result
TAM	~40–60k enterprises worldwide will run >=3 production genAI use cases by 2028; governance layer value US\$30–60k ACV	US\$1.5–3.5B annual [model]
SAM (initial)	Regulated verticals + GCCs in India/EU/US reachable via open-core + SI channels: ~6–8k accounts	US\$250–450M annual [model]
SOM (36 months)	120–180 paying accounts at US\$35–50k blended ACV via design-partner -> partner-led motion	US\$5–8M ARR [model]

7.3 · Competitive positioning

Player (representative)	Strength	Gap ControlPlane occupies
Portkey / Kong / LiteLLM (gateways)	Routing, caching, cost visibility, scale	Thin decision layer; no per-pipeline verdict governance; no audit-grade evidence
Bedrock Guardrails / Azure Content Safety / NeMo	Distribution, harm filters	Cloud-locked or single-plane; one global config; no fail-posture or borderline policy; no HITL lineage
Arize / Langfuse / LangSmith / Datadog	Observability depth, enterprise reach	Post-hoc by design — cannot gate delivery; no enforcement
Lakera / prompt-security startups	Focused attack research	Single plane (security); no cost/quality convergence, no per-use-case governance
Presidio / Google DLP	PII category breadth	Not a delivery-path decision system; we integrate rather than compete (roadmap: Presidio as a tier-1 provider)

We state plainly what our own analysis discipline required: every individual technique exists somewhere. The defensible position is the **converged, per-use-case, evidence-producing decision layer** — and a credibility asset none of the above ships: auditable claims about their own accuracy.

8 · Business model and go-to-market

8.1 · Open-core model

Tier	What ships	Price posture [model]
Open source (Apache-2.0)	Gateway, policy engine, detectors, audit schema, eval harness — the full prototype lineage	Free — adoption, community patterns, credibility
Team	Governance Console (the dashboard shipped in this prototype, hardened), SSO, retention, alerting	US\$500–1,000 / month per production pipeline
Enterprise	Compliance policy packs (EU AI Act, DPDP, RBI/SEBI, HIPAA), audit exports, air-gapped/local-model deployment, support SLAs, custom calibration on customer traffic	US\$40–120k ACV

Per-pipeline pricing scales with the customer's own AI adoption — land on one use case, expand as the platform team routes more pipelines through the plane. Local-model support (already working: the demo's fallback is a fully local Ollama deployment) unlocks the air-gapped segment cloud guardrails cannot serve.

8.2 · Go-to-market phases

Phase	Motion	Proof point targeted
Launch (0–3 mo)	Public OSS release + the build-in-public story (the adjudication ledger and blind-measurement discipline are genuinely distinctive developer content); Show-HN/launch posts; conference CFPs on 'AI built under AI oversight'	1k GitHub stars-class attention; inbound design-partner pipeline
Design partners (2–6 mo)	3–5 pilots in Indian BFSI/GCC + one EU-exposed SaaS; success = one pipeline in production with audit evidence accepted by the partner's risk function	Reference architecture + first case studies; calibration-on-customer-traffic playbook
Partner-led (6–18 mo)	System-integrator channel — the governance layer is precisely what SIs (Accenture among them) deploy for clients; cloud-marketplace listings	Repeatable SI-attached deals; first US\$1M ARR [model]

The Accenture channel is not rhetorical: this proposal exists inside Accenture's own innovation program, and responsible-AI deployment is a stated Accenture service line. A winning prototype here is a warm first partnership conversation.

9 · Impact

Dimension	Mechanism	Evidence base
Privacy protection	PII intercepted at input (provider never receives it) and output; credentials blocked as incidents; zero raw values in evidence — verified by an owner-run audit-DB hunt returning 0	Live prototype behavior; NFR-SEC-001 gate
Regulatory de-risking	Append-only audit trail, HITL lineage, per-jurisdiction policy packs -> evidence for DPDP (up to INR 250 crore exposure) and EU AI Act oversight duties	Shipped audit schema + review queue
Cost discipline	Pre-dispatch budget enforcement, loop guards, two-tier routing; savings scale with the deployment's price ratio (measured 2.0× on demo tiers; flagship-vs-mini gaps run far higher)	Relative-delta simulation, honestly labeled
Safe AI adoption at scale	Small teams get bank-grade oversight from a config file — lowering the barrier for Indian enterprises and public-sector pilots to deploy AI responsibly	One-URL integration; local-model path
Engineering practice	The agent-oversight methodology (contract, deviation protocol, independent review) is publishable and reusable beyond this product	Public ledger in the repository

10 · Phased roadmap

Phase 0 is complete and is this prototype. Later phases burn down the standing-limitation register — our roadmap is a list of measured gaps, not imagined features.

Phase	Scope	Exit criteria
P0 · Prototype (done)	Gateway, tri-plane detectors, policy engine, audit + HITL, console, demo; evidence pipeline	This submission
P1 · Pilot-ready (0–3 mo)	Recall categories & Indic/locale PII; threshold sweeps via the feedback loop on partner traffic; fast_consistency & deep-audit workers; Presidio head-to-head on the frozen corpus; conversation-level cumulative risk	Recall ≥ 0.95 on expanded corpus or an explained delta; 3 pilots live
P2 · Product (3–9 mo)	Governance Console GA; compliance policy packs; compiled data plane (Rust) for the microsecond story at scale; multi-tenant isolation; SOC 2 program start	First paid conversions; P99 targets at 10× load
P3 · Scale (9–18 mo)	Kubernetes sidecar/mesh deployment; agent-action gating (tool-call firewall); marketplace listings; policy-pack certification with partners	Partner-led revenue; US\$1M ARR [model]

11 · Key risks and mitigations

Risk	Reality check	Mitigation
Detection quality gaps (our blind numbers show them)	Injection recall 0.150 blind; calibration band inverted on synthetic corpus	Layered defense (patterns already catch overt attacks); feedback loop exists precisely to tune thresholds on real traffic; per-customer calibration is the paid motion — and publishing these numbers is why a buyer can trust the ones that improve
Hyperscaler bundling	AWS/Azure/GCP ship guardrails by default	Cross-provider + local-model neutrality; governance depth (per-pipeline verdicts, fail posture, HITL lineage) that platform filters don't attempt; compliance-evidence moat
Latency objections	Real: one detector tail-breaches its budget (25.35 vs 25 ms), published	Sub-ms deterministic tier already measured; attribution-correct enforcement; compiled data plane in P2; per-pipeline max-token policy as the pressure valve
Provider churn	A model we depended on was deprecated 11 days before this deadline	It broke loudly, not silently: provenance classes + boot canary are shipped features born from this exact event
Student team capacity	Three third-years	The delivery record in the public ledger — measured discipline most professional teams cannot evidence; funding converts our documented process into hired capacity; SI channel supplies deployment muscle
Synthetic-corpus overfit claims	Fair critique of any self-built eval	Corpus frozen before tuning; blind-first protocol; Presidio head-to-head and partner-traffic calibration scheduled in P1

12 · The ask

From this challenge: the win and the platform it provides — Accenture mentorship and a first design-partner introduction inside Accenture's responsible-AI practice.

Indicative pre-seed [model]: INR 1.6 crore (~US\$190k) for 12 months — 60% founding-team + one senior engineer; 20% pilot infrastructure and compliance advisory (DPDP/AI-Act policy-pack review); 20% GTM content and developer relations. Milestones: 3 pilots live (Q2), recall/threshold targets met on partner traffic (Q3), first paid conversions and SOC 2 kickoff (Q4).

Every company is deploying AI faster than it can supervise it. We built the supervision — configurable per business context, honest about its numbers, and proven by the way it was built. ControlPlane: oversight for AI, built under oversight.

Team b24bb1029 · Priyanshu Pandey · Harshit Saini · Jayant Soni — Indian Institute of Technology Jodhpur. Public repository, demo video, and README accompany this proposal per Round 2 requirements.